Narges Rezaie

Data Scientist, Bioinformatics Engineer, Computational Biologist, Data Analyst, AI/ML Engineer

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Summary

Bioinformatics engineer and computational biologist with a Ph.D. in Mathematical, Computational, and Systems Biology from UC Irvine and 7+ years of experience, including 2 years of post-PhD industry experience, developing and deploying scalable bioinformatics pipelines for genomic, sequencing, and biomedical data. Experienced in long-read and short-read sequencing applications, including genome assembly, variant calling, metagenomics, and RNA-seq, with strong expertise in workflow orchestration using Nextflow and Snakemake, benchmarking, QC, and productionizing bioinformatics tools in AWS and HPC environments. Proficient in Bash, Python, R, and C/C++, with a strong foundation in statistics, machine learning, reproducible software development, and using LLM- and agent-based systems to support bioinformatics workflows and internal tooling.

Work Experience

Bioinformatics Pipeline Engineer — **CA, USA**
Plasmidsaurus — *Dec 2024 – Present*

- Owned the end-to-end development, productionization, and maintenance of bioinformatics pipelines for sequencing-based products, with primary ownership of bacterial production workflows.
- Led bacterial pipeline development from method evaluation and workflow design through validation, deployment, troubleshooting, and final customer-facing deliverables.
- Contributed to RNA-seq pipeline development and production support, including workflow improvements, troubleshooting, and cross-functional collaboration.
- Built scalable and reproducible workflow systems using Nextflow and Snakemake across AWS and high-performance computing environments.
- Benchmarked and integrated bioinformatics tools, reference databases, and QC strategies to improve taxonomic profiling, host-read removal, assembly quality, and variant detection in production workflows.
- Resolved pipeline failures and biological, sequencing, and infrastructure-related edge cases across the production lifecycle, from data ingestion to final result generation.
- Collaborated with scientists, engineers, product, and support teams to translate computational methods into robust internal workflows and customer-facing bioinformatics products.
- Explored and integrated new computational methods, bioinformatics tools, and LLM/agent-based systems to improve workflow automation, analysis, and internal tooling.

Research Experience

Postdoctoral fellow — **CA, USA**
Mortazavi Lab, University of California, Irvine — *March 2024 – Jan 2025*

- Developed, applied, and optimized bioinformatic algorithms, tools, and workflows to analyze NGS datasets including short- and long-read RNA-seq at bulk and single-cell resolution, as well as Multiome, Perturb-seq, and DNA methylation data.
- Characterized the impact of genomic variants, regulatory elements, and genes on molecular and cellular phenotypes by analyzing naturally occurring or designed genomic perturbations across various cellular models (Publication).
- Identified active regulatory elements and genes for individual cell types and states by analyzing single-cell datasets.
- **Gene program evaluation (GitHub)**: Implemented a flexible framework to evaluate the plausibility of gene programs inferred from single-cell genomic data (scRNA-seq, scATAC-seq, Multiome, and Perturb-seq) using computational methods.

Graduate Researcher — **CA, USA**
Mortazavi Lab, University of California, Irvine — *Sept 2019 – March 2024*

- **Model-AD (GitHub)**: Analyzed diverse genomics datasets (bulk, single-cell, and single-nucleus) to characterize next-generation animal models for Alzheimer's disease and develop predictive models for therapeutic discovery. Identified gene and transcript expression changes in Model-AD mice relative to wild-type mice and human Alzheimer's disease patients. Implemented standardized data acquisition to Synapse.

- **IGVF**: Investigated how genomic variation affects genome function and phenotype across tissues using single-cell data.
- **ENCODE4**: Integrated and interpreted single-cell omics data to profile full-length transcriptomes in humans and mice.
- **Gene Regulatory Networks**: Built models of gene regulatory networks using bulk and single-cell RNA-seq and ATAC-seq data in humans, rodents, and other vertebrates to study how developmental logic is encoded in the genome.
- **PyWGCNA (GitHub)**: Developed a Python package for weighted correlation network analysis (WGCNA), enabling comparison of networks across datasets and against external gene lists using Fisher's exact test to assess module functional enrichment.
- **Topyfic (GitHub)**: Developed a Python package to identify gene programs (topics) using a robust unsupervised machine learning approach based on Latent Dirichlet Allocation, and annotated gene programs using internal (trait and clinical information) and external (e.g., Enrichr, REACTOME) resources.
- Benchmarked doublet detection methods on single-nucleus RNA sequencing (snRNA-seq) datasets.
- Worked with experimental scientists to understand and support their research goals.
- Mentored undergraduate students in programming and bioinformatics pipelines.

Research Assistant **Tehran, Iran**
Sharif University of Technology *May 2016 – Aug 2019*

- **Statistical Analysis of Clinical Data - Golestan Cohort (GitHub)**: Designed a statistical model to predict cardiovascular disease risk using data from the Golestan Cohort Study, which investigates environmental and genetic risk factors for esophageal squamous cell carcinoma (ESCC) in Golestan Province, Iran, and includes approximately 50,000 adults aged 40–75 years.
- **SomaGene (GitHub)**: Developed an R package to analyze somatic mutations in cancer and identified breast cancer-specific ncRNA genes with potential relevance to the underlying genetic basis of breast cancer (Publication).
- **MaxHiC (GitHub)**: Developed a machine learning-based tool to identify significant interacting regions from Hi-C and capture Hi-C data (Publication).

Education

PhD in Mathematical, Computational, and Systems Biology **USA**
University of California, Irvine *2019–2024*

- **Thesis**: Computational approach to characterize gene dynamics using bulk and single-nucleus RNA sequencing to study Alzheimer's disease. **Supervisor**: Dr. Ali Mortazavi. **GPA**: 3.98/4.0

B.S. in Information Technology in Computer Engineering **Iran**
Sharif University of Technology *2014–2018*

Skills

- **Bioinformatics:** Genome assembly, small variant calling, metagenomics, bulk RNA-seq, single-cell/single-nucleus RNA-seq, multiome analysis, QC and benchmarking
- **Programming:** Python, R, Bash, SQL, C/C++, Java
- **Python libraries:** pandas, NumPy, SciPy, scikit-learn, PyTorch, TensorFlow, Polars, AnnData, Biopython, Matplotlib, pytest
- **Workflow & pipeline development:** Nextflow, Snakemake, DSL2, workflow optimization, reproducible pipelines
- **Cloud / infrastructure:** AWS, AWS Batch, S3, SLURM, Docker, Git, Linux
- **Bioinformatics tools:** Minimap2, Samtools, bedtools, BLAST, Picard, RSEM, Kallisto, Seurat, Scanpy, Cellbender, Clair3, Flye, Hifiasm, autocycler
- **Machine learning / AI:** Statistical modeling, unsupervised learning, LLM- and agent-based workflow automation
- **Developed by me:** PyWGCNA, Topyfic

Publications

- **N Rezaie**, ..., B Wold, A Mortazavi. Identification of robust cellular programs using reproducible LDA that impact sex-specific disease progression in different genotypes of a mouse model of AD. bioRxiv (2024)

- **N Rezaie**, F Reese, A Mortazavi. PyWGCNA: A Python package for weighted gene co-expression network analysis. Bioinformatics (2023)
- **N Rezaie**, ..., H Alinejad-Rokny. Somatic point mutations are enriched in non-coding RNAs with possible regulatory function in breast cancer. Communications Biology (2022)
- M Bayati*, **N Rezaie*** and M Hamidi, MS Tahaei, H Rabiee. A New R Package for Categorizing Coding and Non-Coding Genes. Preprints (2023)
- IGVF Consortium. Deciphering the impact of genomic variation on function. Nature (2024)
- RK Loving, ..., **N Rezaie**, ..., A Mortazavi, L Pachter. Long-read sequencing transcriptome quantification with lr-kallisto. bioRxiv (2024)
- E Rebboah, **N Rezaie**, ..., B Wold, A Mortazavi. The ENCODE mouse postnatal developmental time course identifies regulatory programs of cell types and cell states. bioRxiv (2024)
- LF Garcia-Agudo, ..., **N Rezaie**, ..., GR MacGregor, KN Green. BIN1^{K358R} suppresses glial response to plaques in mouse model of Alzheimer's disease. Alzheimer's & Dementia (2024)
- CA Butler, ..., **N Rezaie**, ..., GR MacGregor, KN Green. The Abca7^{V1613M} variant reduces Aβ generation, plaque load, and neuronal damage Alzheimer's & Dementia (2024)
- F Reese, ..., **N Rezaie**, ..., BJ Wold, A Mortazavi. The ENCODE4 long-read RNA-seq collection reveals distinct classes of transcript structure diversity. bioRxiv (2023)
- KM Tran, ..., **N Rezaie**, ..., GR MacGregor, KN Green. A Trem2^{R47H} mouse model without cryptic splicing drives age-and disease-dependent tissue damage and synaptic loss in response to plaques. Molecular neurodegeneration (2023)
- H Alinejad-Rokny, ..., **N Rezaie**, KT Tam, ARR Forrest. MaxHiC: A robust background correction model to identify biologically relevant chromatin interactions in Hi-C and capture Hi-C experiments. PLOS Computational Biology (2022)
- **N Rezaie**, ..., H Alinejad-Rokny. Somatic point mutations are enriched in non-coding RNAs with possible regulatory function in breast cancer. Communications Biology (2022)
- DI Javonillo, ..., **N Rezaie**, ..., KN Green, FM LaFerla. Systematic phenotyping and characterization of the 3xTg-AD mouse model of Alzheimer's Disease. Frontiers in neuroscience (2022)
- S Forner, ..., **N Rezaie**, ..., FM LaFerla, KN Green. Systematic phenotyping and characterization of the 5xFAD mouse model of Alzheimer's disease. Scientific data (2021)
- G Balderrama-Gutierrez, H Liang, **N Rezaie**, ..., F LaFerla, A Mortazavi. Single-cell and nucleus RNA-seq in a mouse model of AD reveal activation of distinct glial subpopulations in the presence of plaques and tangles. bioRxiv (2021)

Selected Talks and Posters

- Alzheimer's Association International Conference 2024. *Poster*
  5xFAD/Abi3^{S209F} mice display distinct age-dependent effects on amyloid beta pathology and microglia dynamics.
- Alzheimer's and Brain Awareness Month, PARSE biosciences 2023. *Invite speaker*
  Identifying Distinct Cellular Programs from Single Cell Datasets Using Topyfic.
- Network Biology 2023 (CHSL) and Probabilistic Modeling in Genomics 2023 (CHSL). *Poster*
  Topyfic identified robust cellular programs using reproducible Latent Dirichlet allocation (rLDA).
- Alzheimer's & Dementia 2022 (AAIC). *Poster*
  Bulk and single-cell gene expression network analysis in mouse models of AD using PyWGCNA.
- Alzheimer's & Dementia 2021 (AAIC)). *Poster*
  The MODEL-AD Explorer: An open-access resource for comparing phenotypic data from mouse models of Alzheimer's disease.
- Neuroscience 2020 (SFN) *Poster*
  Bulk and single-nucleus analysis of the 3xTgAD mouse model.

References

Available upon request.